ex de pe curs

1. $x^{(1)} = 3 \quad x^{(2)} = 1 \quad x^{(3)} = -1$

$y^{(1)} = 1 \quad y^{(2)} = 1 \quad y^{(3)} = -1$

$\max_{w, w_0, h} h$

subject to $y^{(i)}(w^T x^{(i)} + w_0) \geq h \quad 1 \leq i \leq m$

$\|w\|_2 = 1 \quad , h > 0$

$$\begin{cases} y^{(1)}(w^T x^{(1)} + w_0) \geq h \\ y^{(2)}(w^T x^{(2)} + w_0) \geq h \\ y^{(3)}(w^T x^{(3)} + w_0) \geq h \end{cases}$$

$$\begin{cases} 1(3w + w_0) \geq h \\ 1(1w + w_0) \geq h \\ -1(-1w + w_0) \geq h \end{cases} \Rightarrow \begin{cases} 3w + w_0 \geq h \\ w + w_0 \geq h \quad (1) \\ w - w_0 \geq h \quad (2) \end{cases}$$

$(1)+(2) \Leftrightarrow 2w \geq 2h \Leftrightarrow w \geq h$

Vrem să maximizăm $h \Rightarrow h = w$

$$\Rightarrow \begin{cases} 3w + w_0 \geq w \\ w + w_0 \geq w \\ w - w_0 \geq w \end{cases} \Leftrightarrow \begin{cases} 2w \geq -w_0 \\ w_0 \geq 0 \quad (3) \\ 0 \geq w_0 \quad (4) \end{cases}$$

din (3), (4) $\Rightarrow \quad w_0 \geq 0 \geq w_0 \Rightarrow \boxed{w_0 = 0}$

$\Rightarrow \quad 2w \geq 0 \Rightarrow \boxed{w \geq 0}$

Din formulă $\|w\|_2 = 1 \Rightarrow \boxed{w = 1}$

$\|w\|_2 \rightarrow \sqrt{\sum_{j=0}^{d} |w_j|^2}$

2.

$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)})\} = \{(1,1), (2,-1), (3,2)\}$, $\eta = 0.1$, $\lambda = 1$, $r = 0.5$.

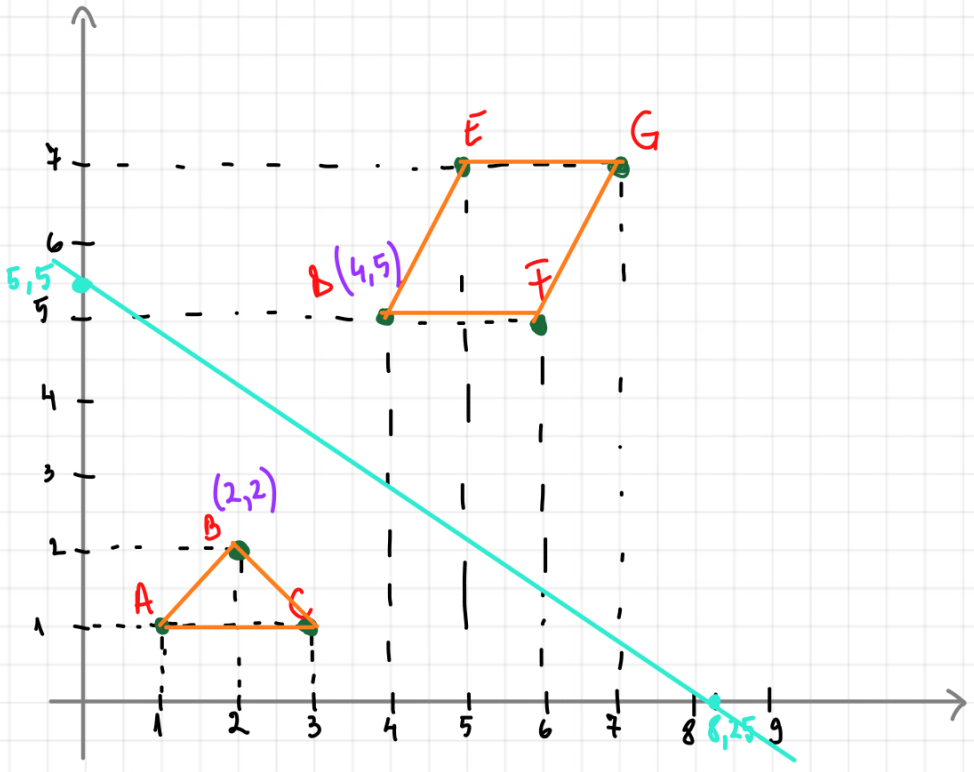
2. Assume we have a training set for two-class classification in two dimensions that contains seven sample points: points $(1,1), (2,2), (3,1)$ with label $+1$ and points $(4,5), (5,7), (6,5), (7,7)$ with label -1. Which points are the support vectors for a hard-margin SVM?

[1 3]

$y_1 = y_2 = y_3 = 1$

$y_4 = y_5 = y_6 = y_7 = -1$

$\min \frac{1}{2} \|w\|_2^2 \qquad y^{(i)} (w^T x^{(i)} + w_0) \geq 1 \quad , \quad 1 \leq i \leq m$



Marginea este minimul distanței dintre figuri.

Este ușor de observat că $B(2,2)$ și $B(4,5)$ sunt cele mai apropiate.

$$d(B,B) = \sqrt{(4-2)^2 + (5-2)^2} = \sqrt{4+9} = \sqrt{13}$$

Marginile pot fi descrise ca linii paralele, deci avem ecuațiile:

$$\begin{cases} y = ax + b \\ y = ax + c \end{cases} \Leftrightarrow \begin{cases} 2 = 2a + b \\ 5 = 4a + c \end{cases} \Rightarrow \begin{cases} b = 2 - 2a \\ c = 5 - 4a \end{cases}$$

Ecuațiile devin

$$\begin{cases} y = ax + (2-2a) \\ y = ax + (5-4a) \end{cases} \quad \oplus$$

$\Leftrightarrow 2y = 2ax + 7 - 6a$

$\Leftrightarrow y = ax + \frac{7-6a}{2}$

Formula distanței dintre 2 drepte: $\frac{|c-b|}{\sqrt{1+a^2}}$

$\Rightarrow \frac{|5-4a-2+2a|}{\sqrt{1+a^2}} = \frac{|3-2a|}{\sqrt{1+a^2}}$

Pp. $3-2a \geq 0 \Rightarrow \frac{3-2a}{\sqrt{1+a^2}} = \sqrt{13} \quad \uparrow^2$

$\frac{9-12a+4a^2}{1+a^2} = 13$

$\Rightarrow 9-12a+4a^2 = 13+13a^2$

$9a^2 + 12a + 4 = 0$, $\Delta = 144 - 4\cdot 4\cdot 9 = 144 - 16\cdot 9 = 0$

$\Rightarrow a_1 = a_2 = \frac{-12}{2\cdot 9} = -\frac{2}{3}$

$\Rightarrow y = -\frac{2}{3}x + \frac{7-6(-\frac{2}{3})}{2} = -\frac{2}{3}x + \frac{11}{2}$

Dacă dreapta împarte punctele, atunci B și B sunt vectori - suport

$x = 0 \Rightarrow y = \frac{11}{2} = 5{,}5$

$y = 0 \Rightarrow \frac{2}{3}x = \frac{11}{2} \Rightarrow x = \frac{33}{4} = 8{,}25$

$\Rightarrow$ împarte punctele deci este bine

$m_{BC} = \frac{1-2}{3-2} = -1$
$m_d = -\frac{2}{3}$
$\Rightarrow C \not\in v_{sup}$

③. $\{(1,1), (1,2), (2,1), (2,2), (5,5), (5,6), (6,5), (6,6)\}$

$x_t = (3,3)$

$y_1 = y_2 = y_3 = y_4 = 1$

$y_5 = y_6 = y_7 = y_8 = -1$

pt $K=1$, $y_t = ?$ K-nearest neighbors

$d(x_1, x_t) = \sqrt{(3-1)^2 + (3-1)^2} = \sqrt{8}$

$d(x_2, x_t) = \sqrt{5}$

$d(x_3, x_t) = \sqrt{5}$

$d(x_4, x_t) = \sqrt{2}$

$d(x_5, x_t) = \sqrt{8}$

$d(x_6, x_t) = \sqrt{13}$

$d(x_7, x_t) = \sqrt{13}$

$d(x_8, x_t) = \sqrt{18}$

Ordine: $x_4, x_2, x_3, x_1, x_5, x_6, x_7, x_8$

$y_t = y_4 = +1$

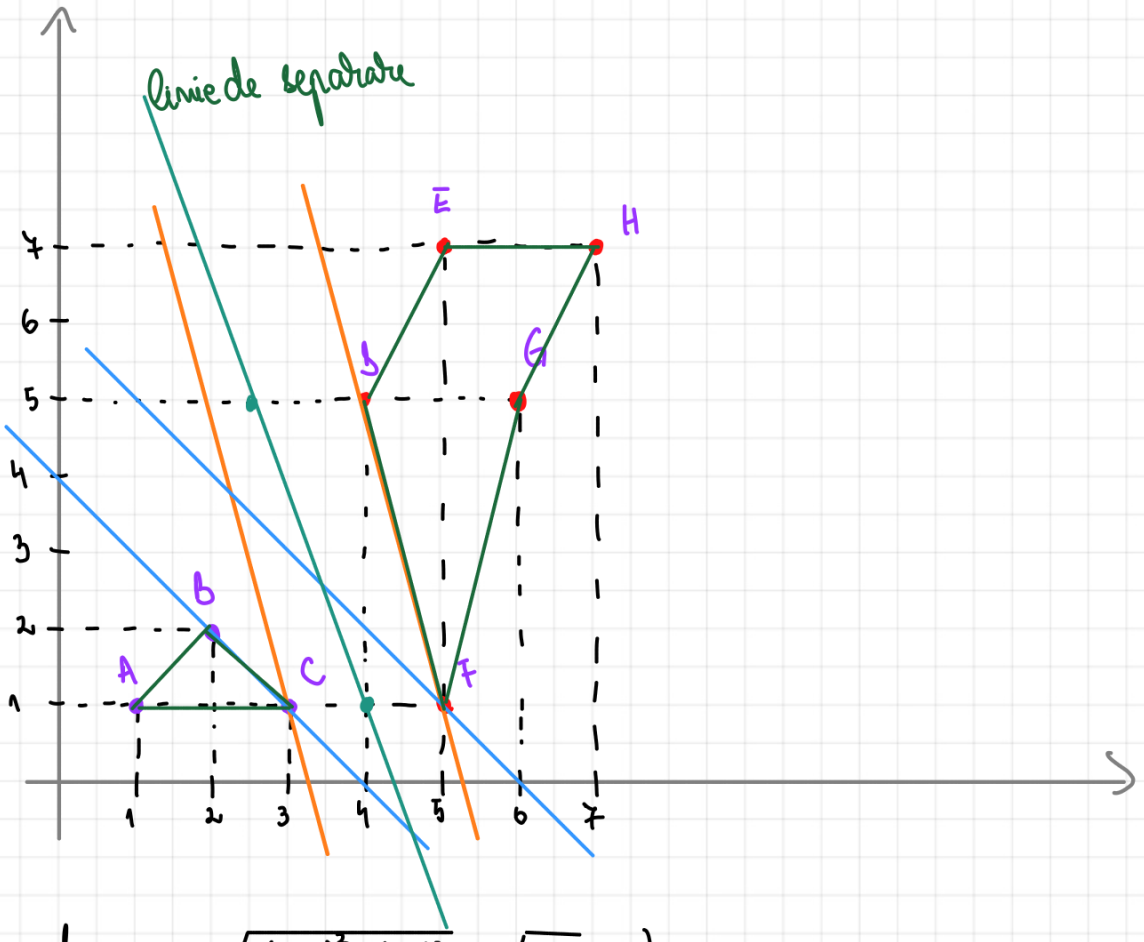
$k=3 \Rightarrow \begin{cases} y_4 = +1 \\ y_2 = +1 \\ y_3 = +1 \end{cases} \Rightarrow y_t = +1$

$k=5 \Rightarrow \begin{cases} y_4 = +1 \\ y_2 = +1 \\ y_3 = +1 \\ y_1 = +1 \\ y_3 = -1 \end{cases} \Rightarrow y_t = +1$

sau $d(x_1, x_t) = |3-1| + |3-1| = 4$

(4.)

2. Assume we have a training set for two-class classification in two dimensions that contains eight sample points: points $(1,1),(2,2),(3,1)$ with label $+1$ and points $(4,5),(5,7),(5,1),(6,5),(7,7)$ with label -1. Which points are the support vectors for a hard-margin SVM? Motivate your answer.



$d_{BF} = \sqrt{(5-2)^2+(1-2)^2} = \sqrt{10}$

$d_{CB} = \sqrt{(4-3)^2+(5-1)^2} = \sqrt{17}$

$\Rightarrow$ alegem CB pt că maximizează distanța dintre cele mai apropiate puncte

$d_{CF} = \sqrt{(5-3)^2+(1-1)^2} = \sqrt{4}$ sunt pe același y deci nu merg

Putem considera dreapta care trece prin B, F

$$m = \frac{y_F - y_B}{x_F - x_B} = \frac{1-5}{5-4} = -\frac{4}{1} = -4$$

$$y - y_F = m(x - x_F) \Rightarrow y - 1 = -4(x-5)$$

$$\Leftrightarrow y = -4x + 21 \quad \text{pt BF}$$

Dreapta paralelă cu BF care trece prin C

$m = -4$; $y = mx + b \Rightarrow 1 = -4 \cdot 3 + b$

$$\Leftrightarrow b = 13$$

$$\Rightarrow y = -4x + 13$$

$$\begin{cases} y = -4x + 21 \\ y = -4x + 13 \end{cases} \oplus$$

$y = -4x + 17$ linie de separare

$x = 4 \quad , y = 1$

$x = 3 \quad , y = 5$

5.

on
on algorithm:
niformly at rar
g an instance
D(

Assume we have a training set for two-class classification in one dimension that contains three sample points: point $x^{(1)} = 3$ with label $y^{(1)} = 1$, point $x^{(2)} = 1$ with label $y^{(2)} = 1$, and point $x^{(3)} = -1$ with label $y^{(3)} = -1$. What are the values of w and b given by a hard-margin SVM?

La fel ca aia de pe curs (1.)

support vectors -1 și 1

separate de $y = 1 \cdot x + 0$ $w = 1$, $b = 0$ e mai sus rezolvare